Subject - Cryptography and Network Security
Topic - Phishing Website & URL Classification

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Phishing Website Classification with Linear Discriminant Analysis

Executive summary

We built a machine-learning project that classifies precomputed website indicators as phishing or legitimate using Linear Discriminant Analysis (LDA). The work includes a dataset audit, training pipeline, model selection, held-out evaluation, saved outputs and a FastAPI demonstration interface. The local dataset contains 11,054 rows and 30 predictors after excluding the sequential Index and target column.

The grouped evaluation reports 92.90% test accuracy, 91.94% F1 and 0.9758 ROC AUC on 2,210 test records. Identical predictor vectors are kept in one group across the holdout and training cross-validation. The earlier row-wise split reported 91.68% accuracy; the score difference comes from different test membership and evaluation protocol, not a model improvement.

This project provides an inspectable baseline on a fixed benchmark. It does not establish performance on future sites, unseen domains or a live feature-extraction pipeline.

At a glance

Area	Current project state
Task	Binary classification of structured phishing website indicators
Data	11,054 records; 30 predictors; 4,897 phishing and 6,157 legitimate
Method	Classification LDA; SVD and LSQR with automatic shrinkage
Selection	Grouped five-fold training-only CV using F1
Evaluation	First fold of stratified grouped five-fold split; seed 42
Interface	FastAPI localhost app with dataset, methodology, evaluation and prediction

1 Project scope and dataset

The project asks whether a transparent linear classifier can separate phishing from legitimate examples using encoded properties already present in a benchmark dataset. It is a structured-feature task. The implementation does not fetch URLs, crawl pages, resolve domains or derive indicators from live traffic.

Dataset audit

The configured input, data/phishing.csv, has 11,054 rows and 32 columns. Thirty columns are predictors; Index is a sequential row identifier and class is the target. Original label -1 denotes phishing and +1 denotes legitimate. Phishing is mapped to positive class 1 and legitimate to 0.

Audit item	Finding
Rows and columns	11,054 and 32
Predictors	30 encoded numeric indicators
Phishing / legitimate	4,897 / 6,157 (44.3% / 55.7%)
Missing values	None
Full-row duplicates including Index	None
Repeated feature-and-label rows ignoring Index	5,205
Unique predictor vectors	5,785
Vectors with conflicting labels	64
Identifier	Index; 11,054 unique values, excluded
Raw URL or domain key	Not present

Features represent coded URL, page, domain and reputation properties, not raw URL text. Numeric standardization is fitted inside the model pipeline. No imputation is used because the audited file has no missing values.

Duplicate rows and split design

The sequential Index makes complete rows look unique even when their predictors and labels repeat. Ignoring Index reveals 5,205 repeated feature-and-label rows. The earlier row-wise random split allowed identical predictor vectors into both training and test partitions. The current split groups by the full predictor vector, including vectors observed with conflicting labels, and keeps each group in one partition.

Grouping addresses duplicate-vector leakage. It cannot group distinct vectors by domain because the dataset has no URL or domain key, and it does not test future-time generalization.

2 Method and experiment design

The classifier is scikit-learn classification LDA. It models class-conditional feature vectors with Gaussian distributions and a shared covariance matrix. Under this assumption, class score differences are linear. LDA permits correlated predictors, but its linear boundary may not represent nonlinear interactions.

Grouped split and selection sequence

- Load and validate the CSV and expected target labels.
- Map phishing (-1) to positive class 1; remove class and Index from the predictors.
- Group identical predictor vectors, including vectors observed with conflicting labels.
- Use StratifiedGroupKFold with five folds, shuffling and seed 42. The first fold is the approximately 80/20 holdout; groups stay intact.
- Compare SVD LDA and LSQR with automatic shrinkage on training rows only, using grouped five-fold CV F1.
- Fit the selected pipeline on all training rows and evaluate it once on the held-out fold.

The split contains 8,844 training rows and 2,210 test rows. There are 4,628 unique predictor groups in training and 1,157 in testing, with no groups shared across the boundary. Each validation fold fits its own scaling parameters inside the pipeline.

Candidate solvers

SVD is rank-aware and does not support shrinkage. LSQR supports automatic covariance shrinkage, which can stabilize estimates when predictors are correlated. The candidates tie on grouped CV F1 to the precision reported, so SVD is selected by configured candidate order; this is not evidence that it is superior.

Mathematical interpretation

For class k, LDA assigns delta_k(x) = x^T Sigma^-1 mu_k - 1/2 mu_k^T Sigma^-1 mu_k + log(pi_k), where mu_k is the class mean, Sigma is shared covariance and pi_k is the prior. The predicted class has the larger score. Binary between-class scatter has rank at most one, so there is at most one nonzero Fisher discriminant direction. Classification LDA is distinct from using LDA only to reduce dimensions before another classifier.

3 Software implementation

The repository separates data handling, preprocessing, training, prediction and web presentation. The FastAPI application reads saved artifacts; page loads do not retrain the model.

Component	Role
src/data_loader.py	Loads configuration and CSV; validates target labels.
src/preprocessing.py	Maps phishing to positive class, removes target and identifier, and builds numeric scaling.
src/train.py	Groups identical vectors, creates holdout and CV folds, selects and fits LDA, and writes results.
src/predict.py	Loads the saved bundle, checks feature names, and returns class and probability estimates.
app/main.py	FastAPI routes for dataset, evaluation pages and predictions.
app/templates/evaluation.html	Displays saved metrics and explains the grouped split.
config/experiment_config.json	Dataset path, target, seed and fold count.
outputs/ and models/	Generated metrics, comparisons, coefficients, plots and fitted model.

Training outputs

Training writes models/lda_pipeline.joblib, outputs/results.json, results.csv, lda_comparison.csv and coefficients.csv. It also creates plots for class distribution, confusion matrix, ROC, precision-recall and candidate comparison. The saved bundle contains the fitted pipeline, expected feature list and positive-label metadata.

Prediction service

The interface includes overview, dataset, methodology, evaluation and prediction pages. The dataset API returns schema, missing-value counts, class totals and a preview. Prediction accepts a dataset row or feature record and returns a class and model probability estimates; calibration has not been established.

Reproduction

The README describes a Windows workflow: install requirements, run python -m src.train, then start the app with uvicorn app.main:app --reload. Open http://127.0.0.1:8000. The evaluation page reads outputs/results.json and reflects the latest generated metrics when the app is refreshed.

4 Experimental results

The selected SVD model was evaluated on a 2,210-record grouped holdout. Phishing is positive throughout. The confusion counts are 1,157 true negatives, 74 false positives, 83 false negatives and 896 true positives.

Metric	Value	Interpretation
Accuracy	92.90%	Share of test predictions correct
Precision	92.37%	Share of phishing alerts that were phishing
Recall	91.52%	Share of phishing examples detected
F1	91.94%	Harmonic mean of precision and recall
Specificity	93.99%	Share of legitimate examples correctly identified
ROC AUC	0.9758	Ranking discrimination across thresholds
PR AUC / average precision	0.9712	Precision-recall summary with phishing positive

Confusion matrix

Actual class	Predicted legitimate	Predicted phishing
Legitimate	1,157 (TN)	74 (FP)
Phishing	83 (FN)	896 (TP)

The 83 false negatives are phishing examples missed at the default decision rule. Acceptability depends on the cost of missed sites versus false alerts. The project does not tune a deployment threshold; AUC measures do not choose an operating threshold.

Grouped cross-validation comparison

Candidate	Mean grouped CV F1	Fold standard deviation
SVD	0.90677	0.00757
LSQR with automatic shrinkage	0.90677	0.00757

The recorded values tie to the displayed precision. The selected SVD candidate appears first in the configured order; do not describe it as having a higher grouped CV score.

Runtime measurements

The local run recorded 0.0923 seconds for fitting and 0.0218 seconds for prediction plus probability estimation over the 2,210 held-out rows. The derived mean is 0.00985 milliseconds per sample and throughput is about 101,562 samples per second. These single local measurements are not a production latency guarantee.

5 Demonstration and user workflow

The localhost interface is a teaching and inspection tool for the saved experiment. It connects the feature data, LDA method, grouped evaluation and predictions. It is not a live website scanner; the model requires a structured feature vector in the expected schema.

Suggested demonstration

- Start the application with uvicorn app.main:app --reload and open the local address.
- Use Overview to explain classification over precomputed indicators.
- Open Dataset to show labels, missingness and why Index is excluded.
- Open Methodology to explain shared covariance, shrinkage and grouped model selection.
- Open Evaluation to review the grouped holdout metrics and confusion matrix.
- Predict a dataset row and clarify that probabilities are model estimates.
- Close with benchmark limits and the evidence needed before deployment.

Interpreting the accuracy difference

The row-wise experiment reported 91.68% accuracy on 2,211 test rows. The grouped holdout reports 92.90% on 2,210 rows. The protocol and test membership changed, so these values are not a before-and-after comparison of model quality. On this fold, the grouped test examples happened to be easier for the fitted classifier; another fold or new data could produce a lower score.

The grouped estimate addresses repeated feature vectors crossing the train/test boundary. It remains one fold of a fixed dataset and cannot establish generalization to unseen time periods or domains.

Operational notes

The app loads metrics from the generated JSON and predictions from the saved bundle. Retraining should update both the model and evaluation artifacts. Feature names and definitions must match. Keep the app local until input validation, security, monitoring and deployment needs are addressed.

6 Limitations and interpretation

What the grouped evaluation establishes

Identical predictor vectors stay in one fold, preventing repeated vectors from appearing on both sides of the holdout or CV boundary. Distinct vectors are not grouped by domain because no URL or domain key is available. The experiment does not test future-time performance, adversarial changes or changes in feature extraction.

The audit found 64 predictor vectors with conflicting labels. A classifier using only these predictors cannot distinguish rows with the same vector and different labels. Grouping keeps those examples together, but label ambiguity still limits attainable performance.

Feature and model limits

Some indicators describe domain, rank, page behavior or external reputation. Their provenance and freshness are not audited. If a feature uses information unavailable at prediction time, benchmark scores may overstate live performance. The input is a fixed snapshot, not raw URL text.

LDA assumes Gaussian class distributions with a shared covariance matrix. The encoded features take only a few discrete values, so the Gaussian assumption is approximate. LDA yields a linear score but may miss nonlinear interactions. Shrinkage can stabilize covariance estimation; it does not remove the model assumptions.

The project reports AUC measures but no calibration analysis, confidence intervals, repeated seeds, subgroup results or cost-based threshold selection. Treat predict_proba values as model estimates, not calibrated risks. Coefficients are descriptive weights, not causal effects.

Reproducibility and deployment boundary

The notebook, website training code, saved pipeline, plots and metrics in the project workspace use grouped evaluation. A previously launched or deployed copy can still show the earlier row-wise result until refreshed or redeployed.

Before deployment claims, evaluate newer data, use a time-based holdout and domain-aware grouping when identifiers are available, confirm all 30 features exist at the decision point, and choose a threshold based on the costs of missed phishing and false alerts. Monitor drift and label feedback.

7 Conclusions and next steps

The project provides an inspectable LDA baseline for the supplied phishing website feature table. It audits labels and predictors, excludes Index, compares two solver configurations, keeps duplicate feature vectors together during grouped holdout and training CV, writes reusable model artifacts, and serves the saved model through a local FastAPI interface.

The grouped holdout reports 92.90% accuracy, 91.94% F1, 91.52% phishing recall and 0.9758 ROC AUC on 2,210 rows. The confusion matrix records 83 missed phishing examples and 74 false alerts. Its higher accuracy than the earlier row-wise 91.68% reflects different test membership and reduced duplicate-vector leakage; it is not evidence that the model improved. These results apply to this grouped fold of the benchmark.

Recommended follow-through

- Use the Colab notebook and grouped website pipeline together; refresh the app before a demonstration.
- Record the split method with every metric and avoid comparing different holdout protocols as if they were the same test.
- Evaluate on newer, time-based data and group by domain when identifiers become available.
- Study threshold selection, calibration, confidence intervals and error costs.
- Document feature provenance, freshness and availability at inference time.
- Keep the web app local until operational security and monitoring are addressed.

Sources and project records

Primary records: README.md; config/experiment_config.json; src/data_loader.py; src/preprocessing.py; src/train.py; src/predict.py; app/main.py; app/templates/evaluation.html; reports/dataset_profile.md; outputs/results.json; outputs/lda_comparison.csv; Phishing_LDA_Colab.ipynb.

Hannousse and Yahiouche, “Phishing website prediction using base and ensemble classifier techniques with cross-validation,” Cybersecurity 5, article 19 (2022), DOI 10.1186/s42400-022-00126-9. This study evaluates LDA; its scores are not directly comparable because its data and protocol differ.

Hannousse and Yahiouche, “Phishing or Not Phishing? A Survey on the Detection of Phishing Websites,” IEEE Access 11 (2023), DOI 10.1109/ACCESS.2023.3247135. Abu Al-Haija and Al Badawi, “URL-based Phishing Websites Detection via Machine Learning,” ICDABI (2021), DOI 10.1109/ICDABI53623.2021.9655851. These works provide field context and are not LDA results for this project.

Project repository: https://github.com/PrasadFakke/phishing-lda